

Marketing Mix Modeling (MMM) — 101

A simple explainer on what is Marketing Mix Modeling (MMM)



ARYMA LABS

AUG 15, 2023



4



Sh

Market Mix Modeling (MMM) is a technique which helps in quantifying the impact of several marketing inputs on sales or Market Share. The purpose of using MMM is to understand how much each marketing input contributes to sales, and how much to spend on each marketing input.

MMM helps in ascertaining the effectiveness of each marketing input in terms of Return on Investment. In other words, a marketing input with higher return on Investment (ROI) is more effective as a medium than a marketing input with a lower ROI.

MMM uses the Regression technique and the analysis performed through Regression is further used for extracting key information/insights.

In this article, I will talk about various concepts associated with understanding MM

1. Multi-Linear Regression:

As mentioned earlier, Market Mix Modeling uses the principle of Multi-Linear Regression. The dependent variable could be Sales or Market Share. The independent variables usually used are Distribution, price, TV spends, outdoor campaigns spend, newspaper and magazine spends, below the line promotional spends, and Consumer promotions information etc. Nowadays, Digital medium is highly used by some

marketers to increase brand awareness. So, inputs like Digital spends, website visits etc. can also be used as inputs for MMM.

An equation is formed between the dependent variables and predictors. This equation could be linear or non-linear depending on the relationship between the dependent variable and various marketing inputs. There are certain variables like TV advertisement which have a non-linear relationship with sales. This means that increase in TV GRP is not directly proportional to the increase in sales. I will discuss about this in more detail in the subsequent section.

The betas generated from Regression analysis, help in quantifying the impact of each of the inputs. Basically, the beta depicts that one unit increase in the input value will increase the sales/profit by Beta units keeping the other marketing inputs constant.

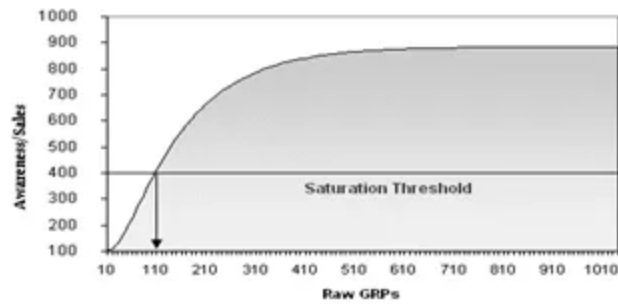
$$\text{Sales} = \beta_0 + \beta_1 * x_1 + \beta_2 * x_2$$

Sales Equation

2. Linear and Non-Linear Impact of predictors:

Certain variables show a linear relationship with Sales. This means as we increase these inputs, sales will keep on increasing. But variables like TV GRP do not have a linear impact on sales. Increase in TV GRPs will increase sales only to a certain extent. Once that saturation point is reached, every incremental unit of GRP would have a negligible impact on sales. So, some transformations are done on such non-linear variables to include them in linear models.

TV GRP is considered as a non-linear variable because, according to marketers an advertisement will create awareness among customers to only a certain extent. Beyond a certain point, increased exposure to advertisement would not create any further incremental awareness among customers as they are already aware of the brand.



So to consider TV GRP as one of the modeling inputs, it is transformed into adstock

TV Adstock has two components.

a. Diminishing Returns: The underlying principle for TV advertisement is that the exposure to TV ads create awareness to a certain extent in the customers' minds. Beyond that, the impact of exposure to ads starts diminishing over time. Each incremental amount of GRP would have a lower effect on Sales or awareness. So, the sales generated from incremental GRP start to diminish and become constant. This effect can be seen in the above graph, where the relationship between TV GRP and sales is non-linear. This type of relationship is captured by taking exponential or log of GRP.

b. Carry over effect or Decay Effect: The impact of past advertisement on present sales is known as Carry over effect. A small component termed as lambda is multiplied with the past month GRP value. This component is also known as Decay effect as the impact of previous months' advertisement decays over time.

$$\text{Adstock}_t = (\text{GRP}_t)^n + (\text{Adstock}_{t-1}) * \lambda$$

3. Base Sales and Incremental Sales:

In Market Mix Modeling sales are divided into 2 components:

a. Base Sales: Base Sales is what marketers get if they do not do any advertisement. It is sales due to brand equity built over the years. Base Sales are usually fixed unless there is some change in economic or environmental factors.

b. Incremental Sales: Sales generated by marketing activities like TV advertisement, print advertisement, and digital spends, promotions etc. Total incremental sales is split into sales from each input to calculate contribution to total sales.

4. Contribution Charts:

Contribution charts are the easiest way to represent sales due to each marketing input. Contribution from each marketing input is product of its beta coefficient and input value.

E.g.: Contribution from Newspaper = β^* Newspaper Spends

To compute contribution %, contribution due to each input is divided by the total contribution. I will elaborate on the interpretation of contribution charts in MMM part 2.

5. Deep Dives

MMM results can be used further to perform deep dive analysis. Deep Dives can be used to assess the effectiveness of each campaign by understanding which campaign or creatives work better than the other ones. It can be used to do a copy analysis of creatives by genre, language, channel etc.

Insights from Deep Dives are considered for Budget optimization. Money is shifted from low performing channels or genres to high performing channels/genres to increase overall sales or market share.



6. Budget Optimization

For any business, Budget optimization is one of the key decisions to be taken for planning purposes.

MMM assists marketers in optimizing future spends and maximizing effectiveness. Using MMM approach, it is established that which mediums are working better than the other ones. Then, budget allocation is done, by shifting money from low ROI mediums to high ROI mediums thus maximizing sales while keeping the budget constant.

So folks, this was a brief about Market Mix Modeling.

Stay tuned for more articles on MMM. Don't forget to subscribe to receive our latest posts on MMM.

For info on how MMM can benefit you, check out the link : [Why Marketing Mix Modeling](#)

For consulting and help with MMM implementation, [Click here](#)

Thanks for reading Aryma Labs' Substack!
Subscribe for free to receive new posts and
support my work.

Subscribe



4 Likes

Discussion about this post

Comments

Restacks



Write a comment...